It's very hard to know the exact composition of a complex organic solid. For example, the chemistry of coal is still not fully understood, despite a long-standing economic incentive. But we've found out some things about Titan tholin. It contains many of the essential building blocks of life on Earth. Indeed, if you drop Titan tholin into water you make a large number of amino acids, the fundamental constituents of proteins, and nucleotide bases also, the building blocks of DNA and RNA. Some of the amino acids so formed are widespread in living things on Earth. Others are of a completely different sort. A rich array of other organic molecules is present also, some relevant to life, some not. During the past four billion years, immense quantities of organic molecules sedimented out of the atmosphere onto the surface of Titan. If it's all deep-frozen and unchanged in the intervening aeons, the amount accumulated should be at least tens of meters (a hundred feet) thick; outside estimates put it at a kilometer deep.

But at 180°C below the freezing point of water, you might very well think that amino acids will never be made. Dropping tholins into water may be relevant to the early Earth, but not, it would seem, to Titan. However, comets and asteroids must on occasion come crashing into the surface of Titan. (The other nearby moons of Saturn show abundant impact craters, and the atmosphere of Titan isn't thick enough to prevent large, high-speed objects from reaching the surface.) Although we've never seen the surface of Titan, planetary scientists nevertheless know something about its composition. The average density of Titan lies between the density of ice and the density of rock. Plausibly it contains both. Ice and rock are abundant on nearby worlds, some of which are made of nearly pure ice. If the surface of Titan is icy, a high-speed cometary impact will temporarily melt the ice. Thompson and I estimate that any given spot on Titan's surface has a better than 50–50 chance of having once been melted, with an average lifetime of the impact melt and slurry of almost a thousand years.

This makes for a very different story. The origin of life on Earth seems to have occurred in oceans and shallow tidepools. Life on Earth is made mainly of water, which plays an essential physical